Lesson 2.4 - Dot Plots and Frequency Tables

> **Video:** Dot Plots and Frequency Tables by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **dot plot** is a picture of a frequency table. Draw a number line from the lowest to the highest value, then stack **one dot per data value** above it. The height of each stack equals the frequency.

Example 1

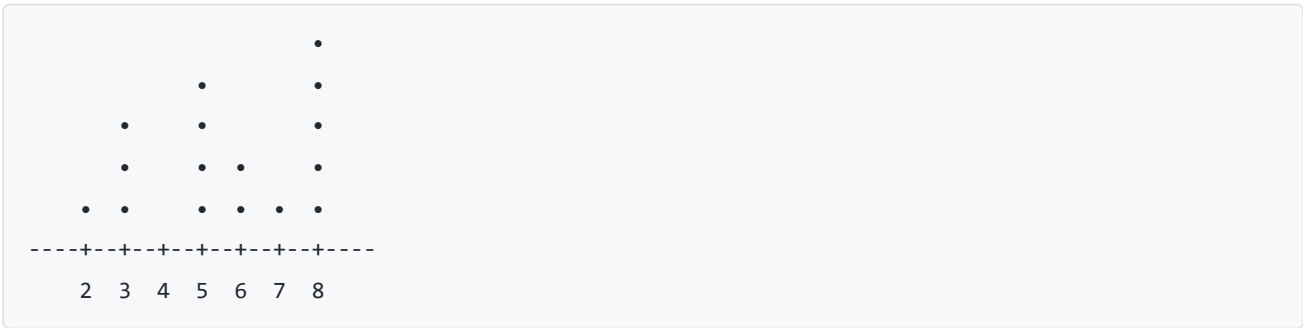
Data (16 values): 5, 8, 3, 6, 8, 5, 8, 2, 3, 8, 5, 8, 3, 5, 7, 6

Step 1 - frequency table

Value	Frequency
2	1
3	3
5	4
6	2
7	1
8	5
Total	**16**

(There is no 4.)

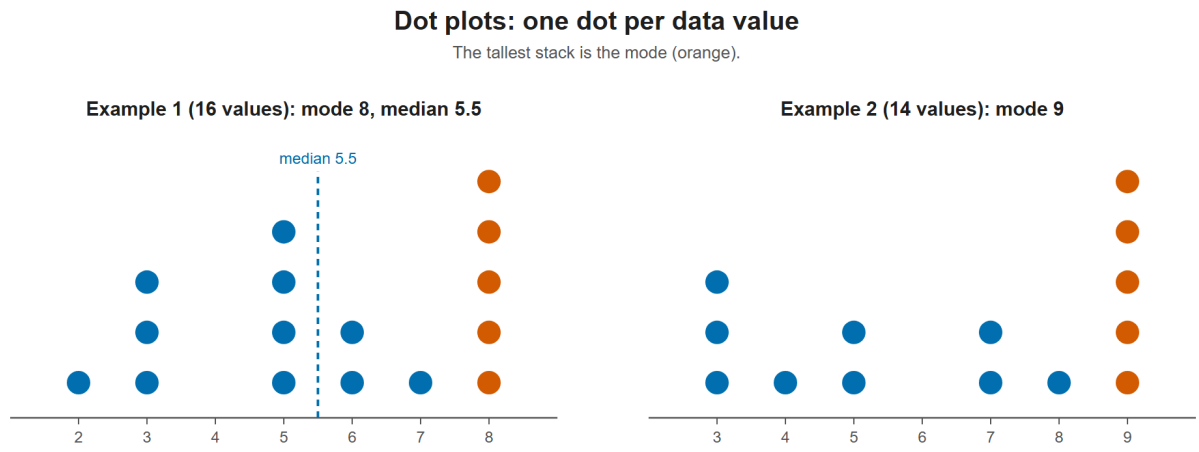
Step 2 - dot plot



Reading the plot:

- **Mode** = the tallest stack = **8**.
- **Median:** with $n = 16$, the median is the average of the 8th and 9th values. Removing dots from both ends leaves a 5 and a 6, so

$$\text{median} = \frac{5 + 6}{2} = 5.5$$



The two dot plots from this lesson. The tallest stack (orange) is the mode. In Example 1 the 8th and 9th values are a 5 and a 6, so the median is 5.5 (dashed line).

Example 2 (practice)

Data (14 values): 7, 4, 9, 3, 9, 5, 9, 7, 3, 9, 3, 9, 5, 8

Counts: 3 → 3, 4 → 1, 5 → 2, 7 → 2, 8 → 1, 9 → 5 (no 6)

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                  •
                  •
    •             •
    •     •     •     •
    •  •  •     •  •  •
----+--+--+--+--+--+--+----
    3  4  5  6  7  8  9
```

Mode = 9 (tallest stack).

Key takeaways

- A dot plot shows **every** data value, so no information is lost.
- Number of dots above a value = its **frequency**.
- **Mode** = tallest column. For the **median**, cross off dots from both ends until the middle is left.